

Clause-Activation Signatures for Attack-Family Attribution in IoMT Using Tsetlin Machines

Rakesh Reddy Yakkati, Per-Arne Andersen, Rishad Shafik, Ole-Christoffer Granmo and Lei Jiao

Abstract—Attack-family attribution in the Internet of Medical Things (IoMT) requires accurate predictions while retaining a model that can be inspected and analyzed. In this paper, we propose Clause-Activation Signatures (CAS), a compact representation of a trained, weighted Tsetlin Machine (WTM) that identifies a small subset of informative clauses while preserving predictive performance. The key idea is to substantially reduce the number of clauses used by the model before performing stability-based selection. Specifically, weight-based pruning first removes clauses with limited contribution, after which complementary-pairs stability selection identifies clauses that are repeatedly selected across trained data. The signs of the averaged clause coefficients are then used to distinguish supporting from opposing evidence. We evaluate CAS on the CICIoMT2024 WiFi/MQTT dataset. The full 2,400-clause TM achieves a macro-F1 of 0.6660 ± 0.0125 , while CAS achieves 0.6660 ± 0.0316 at $\pi_{thr} = 0.8$ and 0.6599 ± 0.0371 at $\pi_{thr} = 0.9$. Including both evidence signs, these settings retain only 216 and 193 distinct clauses on average, respectively, representing a substantial reduction from the original model size with little or no loss in predictive performance. For comparison, five standard classifiers using the same underlying features achieve macro-F1 scores ranging from 0.6284 to 0.6733. These results demonstrate that CAS can substantially simplify a WTM while maintaining competitive performance in attack-family attribution. The resulting compact clause representation reduces the storage footprint for deployment and provides a smaller, more focused set of clauses for subsequent interpretability analysis.

Index Terms—attack attribution, intrusion detection, interpretable machine learning, IoMT, stability selection, Tsetlin Machine

I. INTRODUCTION AND RELATED WORK

Attack-family attribution identifies malicious traffic types in the Internet of Medical Things (IoMT). These labels help analysts interpret security alerts, and this problem has been solved using neural network-based classifiers. For example, Yin et al. [1] used recurrent neural networks to distinguish between normal traffic and four attack categories in the NSL-KDD dataset. To explain predictions, SHAP [2] and LIME [3] interprets post hoc attribute contributions to input features [4]. Deep neural network-based methods are inherently non-explainable, whereas post-hoc methods require additional software applied to an already built non-explainable model. There is a need to develop a model that is inherently

explainable and able to provide analysts with insight by inspecting the detector itself.

The Tsetlin Machine (TM) learns logical conjunctive rules over Boolean literals, which offers a lower memory footprint and interpretable, rule-based learning [5]. Weighted variants associate learned weights with clauses to obtain more compact model representations [6]. Abeyrathna et al. [7] applied TM rules to intrusion detection on KDD’99. Their model provided both predictions and inspectable rules. More recently, Jaiswal et al. [8] evaluated a TM detector on CICIoMT2024 and explained individual predictions through class-wise vote scores and clause-activation heatmaps. Also, in another study [9], the authors used MedSec-25 and examined feature contributions and clause activations. In addition, several datasets, such as CICIDS2017, KDD99, NSL-KDD, UNSW-NB15, and UNSW Bot-IoT, are considered, and anomaly detection is addressed using TM [?]. Since the TM consumes boolean data, [8], [9] and [?] uses a K -bin discretizer from sklearn and a standard booleanizer from the TMU library, respectively. These studies establish both rule-based detection and inspection of the clauses contributing to a prediction. However, previous work has not investigated how to reduce the clause pool while preserving attack-family attribution performance and identifying the most important clauses. Reducing the clause pool can ease the burden of clause-level interpretability analysis and reduce the memory required during inference.

In this paper, we introduce Clause-Activation Signatures (CAS) to select compact, reusable class signatures from a trained clause pool. More specifically, a weight-based pruning first reduces the candidate pool, and it then applies complementary-pairs stability selection (CPSS) [10], [11] to the retained clauses’ activations. Further, averaged coefficients from one-versus-rest models identify supporting and opposing evidence for each class. These evidence signs are estimated separately from the original TM voting signs. CAS scores each flow by normalizing the signed evidence from selected clauses that activate. We evaluate whether CAS reduces the evidence set while retaining predictive performance.

The main contributions include the following aspects:

- A clause-based attribution layer that combines CPSS with signed, normalized evidence to construct reusable class signatures.
- A sufficient margin condition for prediction-preserving TM pruning and an explicit account of the scope of CPSS error control.

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SIGNATURE EXTRACTION

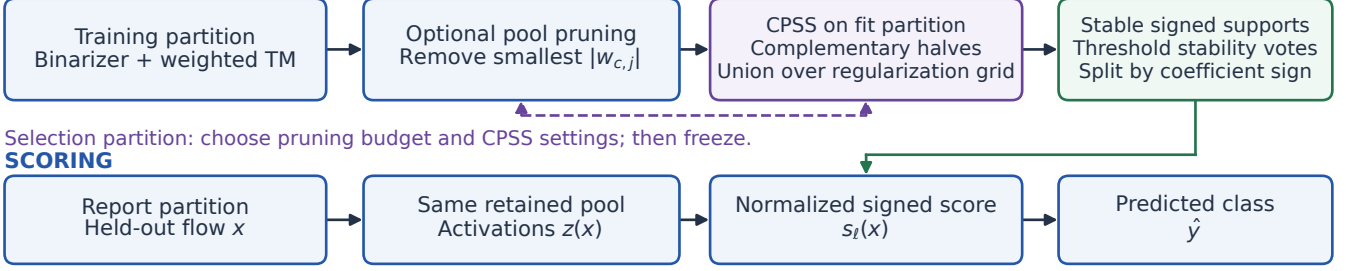


Fig. 1. CAS pipeline with disjoint TM-fit, CPSS-fit, selection, and report partitions.

II. METHOD: CLAUSE-ACTIVATION SIGNATURES

A. Booleanisation and Weighted TM

Figure 1 summarizes the pipeline. Since the TM consumes boolean data, [8], [9] and [?] use a K -bin discretizer from sklearn and a standard booleanizer from the TMU library, respectively. In this paper, training-fitted quantile bins encode numeric features as cumulative thermometer bits (non-finite values are replaced with the finite training maximum).

For N_{cls} classes indexed by $c \in \{1, \dots, N_{\text{cls}}\}$ and C clauses per class indexed by $j \in \{1, \dots, C\}$, the trained pool produces Boolean activations $z_{c,j}(x)$ and signed class sums

$$f_c(x) = \sum_{j=1}^C w_{c,j} z_{c,j}(x), \quad \hat{y}_{\text{TM}}(x) = \arg \max_c f_c(x). \quad (1)$$

Flattening class and clause indices gives the activation matrix $Z \in \{0, 1\}^{n \times p}$ over n samples (flows), initially with $p = N_{\text{cls}}C$.

B. Clause-Pool Pruning

Let $m_{c,j} \in \{0, 1\}$ retain a clause and $f_c(x; m) = \sum_j m_{c,j} w_{c,j} z_{c,j}(x)$. The target is a small pool whose macro-F1 on the selection partition remains within ε of the full model:

$$\min_m \|m\|_0 \quad \text{s.t.} \quad F_{1,\text{sel}}(m) \geq F_{1,\text{sel}}(\mathbf{1}) - \varepsilon. \quad (2)$$

We use a magnitude surrogate instead of solving this discrete objective. For removed indices D_c , set $R_c = \sum_{j \in D_c} |w_{c,j}|$ and $R = \max_c R_c$. Since activations are Boolean,

$$|f_c(x) - f_c(x; m)| \leq R_c \leq R. \quad (3)$$

If $a = \hat{y}_{\text{TM}}(x)$ and $\gamma(x) = f_a(x) - \max_{c \neq a} f_c(x)$, the margin of a over any competing class decreases by at most $2R$. Thus

$$\gamma(x) > 2R \implies \arg \max_c f_c(x; m) = \hat{y}_{\text{TM}}(x). \quad (4)$$

For a fixed deletion budget k per class, the smallest-magnitude weights minimize this surrogate:

$$D_c^*(k) = \arg \min_{|D|=k} \sum_{j \in D} |w_{c,j}|. \quad (5)$$

C. CPSS and Signed Evidence

For each class ℓ , CPSS divides the fit partition into B Complementary pairs as shown in Fig 1. On each of the $2B$ halves, a class-balanced ℓ_1 logistic model predicts the one-versus-rest class label from Z . For half h and inverse-regularisation strength $\kappa \in \Lambda$, let $\beta_{\ell j}^{h,\kappa}$ be the fitted coefficient. A clause receives one vote if any grid point selects it, using numerical tolerance δ :

$$\hat{\pi}_{\ell j} = \frac{1}{2B} \sum_{h=1}^{2B} \mathbf{1} \left\{ \max_{\kappa \in \Lambda} |\beta_{\ell j}^{h,\kappa}| > \delta \right\}. \quad (6)$$

Let $\bar{\beta}_{\ell j}$ average coefficients over all halves and grid points, including zeros. The signed supports are

$$S_{\ell}^{\pm} = \{j : \hat{\pi}_{\ell j} \geq \pi_{\text{thr}}, \quad \pm \bar{\beta}_{\ell j} > 0\}. \quad (7)$$

Positive coefficients provide inculpatory evidence; negative ones provide exculpatory evidence. The attribution rule is

$$s_{\ell}(x) = \frac{\sum_{j \in S_{\ell}^+} \hat{\pi}_{\ell j} z_j(x) - \sum_{j \in S_{\ell}^-} \hat{\pi}_{\ell j} z_j(x)}{\sum_{j \in S_{\ell}^+} \hat{\pi}_{\ell j}}. \quad (8)$$

Prediction uses $\hat{y}(x) = \arg \max_{\ell} s_{\ell}(x)$.

D. Scope of Stability Error Control

For a fixed pool and selector, let p_j denote a clause's population selection probability on a half-sample, $L_{\theta} = \{j : p_j \leq \theta\}$, and q the expected number selected per half. Under the sampling conditions of CPSS [11], the sign-agnostic support S_{ℓ} satisfies, for $\pi_{\text{thr}} > 1/2$,

$$\mathbb{E}|S_{\ell} \cap L_{\theta}| \leq \frac{\theta q}{2\pi_{\text{thr}} - 1}. \quad (9)$$

Taking $\theta = q/p$ yields $q^2/((2\pi_{\text{thr}} - 1)p)$. This controls low-selection-probability inclusions. Interpreting them as false discoveries requires additional assumptions, using $\hat{q} = \sum_j \hat{\pi}_{\ell j}$ is a plug-in diagnostic, not a validated false-discovery certificate. Correlated flows, adaptive operating-point selection, and coefficient signs require separate consideration. No finite bound is claimed at $\pi_{\text{thr}} = 0.5$.

III. EXPERIMENTAL SETUP AND RESULTS

A. Dataset Details

CICIoMT2024 contains 18 attacks on 40 devices, including 25 real and 15 simulated devices [12]. We use its WiFi/MQTT flow features with six labels: Benign, DDoS, DoS, MQTT, Recon, and Spoofing. The selected features exclude IP addresses and port fields. Sampling is capped at 40,000 training and 15,000 test rows per class. The training set contains 195,338 rows (19,291 Benign, 16,047 Spoofing, 40,000 in each remaining class). Meanwhile, the test set contains 80,868 rows (5,868 Spoofing and 15,000 in each other class). Using this data, a stratified split of the training set creates 136,735 TM-fit rows, 39,068 CPSS-fit rows and 19,535 selection rows. The test set serves as the report set for final evaluation. The TM remains frozen during signature extraction, and the CPSS-fit flows are withheld from TM training.

B. TM parameters and hyperparameter optimization

The TM-based booleanizer encodes each feature into bit vectors using quantile-based thresholds. Hence, the 22 numeric features use up to ten quantile bins, giving a cached 154-bit representation that includes redundant bits from repeated quantiles. The TM hyperparameter used in this work is $C = 400$ clauses per class (200 positive and 200 negative), $T = 320$, $s = 8$, and 25 epochs. CPSS uses class-balanced liblinear fits with 200 maximum iterations, $\delta = 10^{-8}$, and $\Lambda = \text{geomspace}(0.001, 0.1, 6)$. Pruning sweeps $k \in \{0, 5, \dots, 395\}$ with $\varepsilon = 0.01$. For each pool, the operating point (B, π_{thr}) is chosen to maximize mean selection macro-F1 over $B \in \{5, 10, 15, 20, 25\}$ and $\pi_{\text{thr}} \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$.

C. Closed-Set Attribution and Sign Validation

Table I compares predictive performance and evidence size. For CAS, size is the mean positive-evidence support summed across classes. It excludes opposing evidence, and a clause selected for multiple classes is counted in each class’s support.

For the pruned pool, the selection-set macro-F1 chooses $(B, \pi_{\text{thr}}) = (10, 0.5)$. This setting achieves report-set macro-F1 0.6755 with 190 positive-evidence clauses. We also highlight $(5, 0.8)$ and $(10, 0.9)$ to illustrate the trade-off at higher selection thresholds and smaller positive supports. At $(5, 0.8)$, CAS uses 149 positive-evidence clauses and matches the full TM’s mean macro-F1 to the reported precision. At $\pi_{\text{thr}} = 0.9$, this count falls to 126 and mean macro-F1 decreases by 0.0061, less than the reported standard deviations across seeds. Including opposing evidence, these two settings use 216 and 193 distinct clauses on average: 11.1–12.4 times fewer than the full pool. Their summed positive/negative supports are 149/199 and 126/173, respectively. Clauses shared across classes explain why the distinct counts are smaller than these sums. The selected $(10, 0.5)$ setting uses 251 distinct signed clauses. All counts are rounded means over three seeds. Figure 2 plots the trade-off using positive-support counts.

Two ablations examine what makes the selected evidence useful. Replacing clause activations with the 154 input literals

TABLE I
REPORT-SET PERFORMANCE (THREE SEEDS). CLAUSE COUNTS DENOTE TM POOL SIZE OR CAS POSITIVE SUPPORTS SUMMED ACROSS CLASSES.

Method	(B, π_{thr})	Clauses	macro-F1	Accuracy
TM, full pool	—	2,400	0.6660 ± 0.0125	0.7033
TM, pruned pool	—	360	0.6615 ± 0.0115	0.6982
CAS, full pool	(10, 0.9)	154	0.6408 ± 0.0168	0.6676
CAS, pruned pool	(10, 0.9)	126	0.6599 ± 0.0371	0.6972
CAS, pruned pool	(5, 0.8)	149	0.6660 ± 0.0316	0.7037
CAS, pruned pool	(10, 0.5)	190	0.6755 ± 0.0183	0.7126
Input literals, same readout	(10, 0.9)	—	0.5876 ± 0.0112	0.6271

TABLE II
STANDARD CLASSIFIERS ON THE 22 NUMERIC FEATURES, THREE-SEED MEANS ON THE REPORT SET.

Model	macro-F1	Accuracy
Random forest [13]	0.6733	0.7251
XGBoost [15]	0.6640	0.7112
Decision tree	0.6526	0.6972
Binarized neural network [14]	0.6503	0.6697
ℓ_1 logistic regression [16]	0.6284	0.6465

under the same readout yields macro-F1 0.5876 ± 0.0112 . This result supports the value of the clauses’ conjunctive structure. A sign-agnostic scorer normalizes the stability-weighted sum over all selected clauses. It yields macro-F1 0.025 ± 0.032 . This drop shows that the readout depends on distinguishing supporting from opposing evidence.

D. Comparison with Standard Classifiers

Table II compares CAS with five standard classifiers. All the baselines are also provided with the same dataset as presented in Section IIIA. We choose hyperparameters on the selection partition and evaluate each model once on the report set. The 300-tree random forest [13] has the highest mean macro-F1 among these baselines, 0.6733. XGBoost follows at 0.6640, the decision tree at 0.6526, the binarized neural network at 0.6503, and the ℓ_1 linear model at 0.6284. Standard deviations across seeds are below 0.002 for the tree models, 0.011 for the linear model, and 0.043 for the BNN. XGBoost gives identical results across seeds in this experiment. It uses no row or column subsampling.

The binarized neural network [14] provides a neural comparison that uses bitwise operations, as does the TM. It is important to note that the BNN reaches 0.6503 using 106,496 binary weights. Meanwhile, CAS attains 0.6660 from 216 distinct signed clauses. These counts describe different model components as they do not establish relative memory or runtime costs.

E. Sensitivity Before and After Pruning

To assess sensitivity to the CPSS settings, we vary the selection threshold and pair count. At $B = 15$, sweeping π_{thr} over $\{0.5, \dots, 0.9\}$ spans macro-F1 0.656–0.674 on the pruned pool and 0.633–0.652 on the full pool. At $\pi_{\text{thr}} = 0.8$, varying B from 5 to 25 gives ranges of 0.656–0.666 and

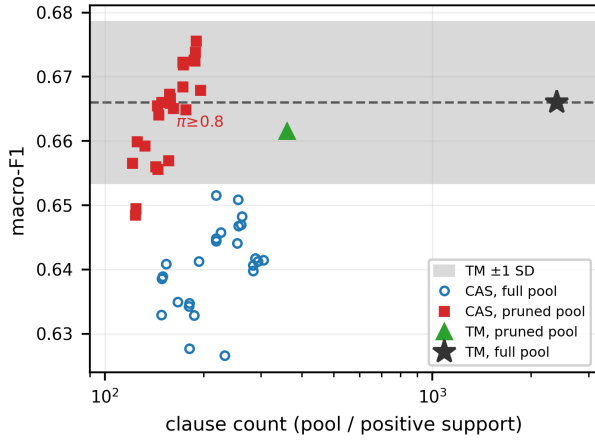


Fig. 2. CAS size-performance trade-off. The band is TM mean $\pm$ SD; 23 of 25 pruned-pool settings lie within it.

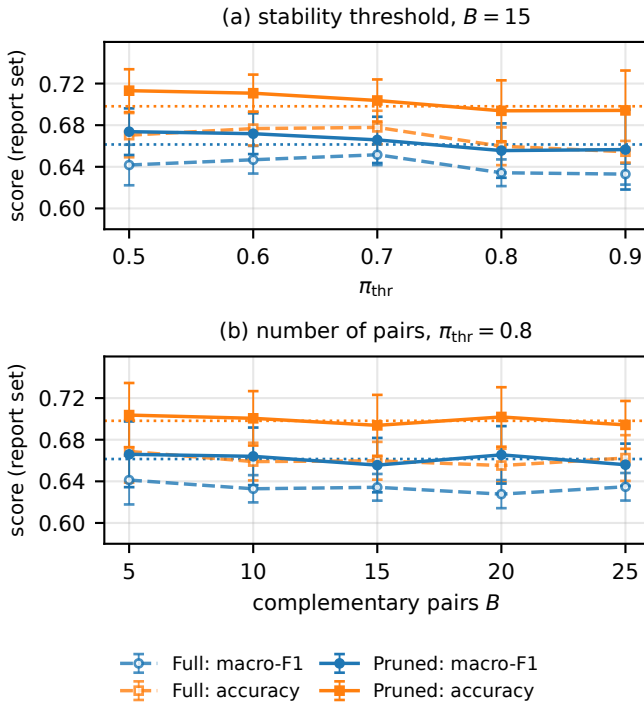


Fig. 3. CAS sensitivity (three-seed mean $\pm$ SD): (a) threshold at $B = 15$; (b) pair count at $\pi_{thr} = 0.8$. Dotted lines: pruned TM.

0.628–0.641, respectively. Fig. 3 compares these sweeps with the pruned-TM baselines. In these sweeps, CAS performs better when selection starts from the pruned pool. This suggests that removing low-weight clauses helps signature extraction in this setting. Across the pruned-pool operating points, positive-evidence support sizes range from 122 to 196 (Fig. 2).

F. Pruning and Signature Size

To examine the effect of pruning alone, Fig. 4 shows TM performance across removal budgets. Report-set macro-F1 and accuracy remain approximately steady as the pool shrinks

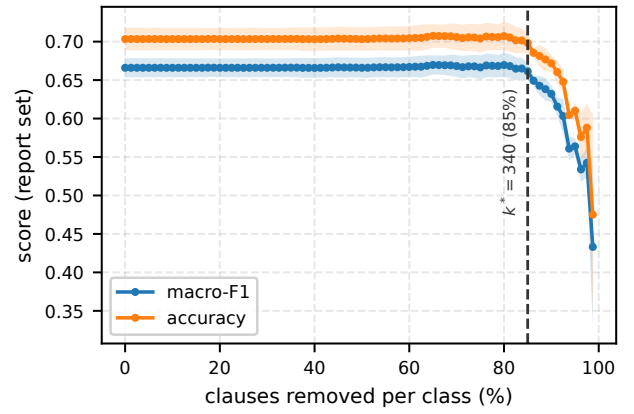


Fig. 4. Pruning sensitivity (mean $\pm$ SD); the dashed line marks $k^* = 340$, chosen on the selection set.

TABLE III
CAS SUPPORTS, SEED 42: FULL (10, 0.9); PRUNED (10, 0.5).

Class	Full pool		Pruned pool	
	$ S^+ $	$ S^- $	$ S^+ $	$ S^- $
Benign	21	65	35	37
DDoS	29	130	29	39
DoS	21	180	38	40
MQTT	21	67	17	20
Recon	19	47	27	54
Spoofing	39	96	39	43
Total	150	585	185	233

from 2,400 to roughly 360 clauses, then decline with further removal. Using only the selection partition to choose the pruning budget gives $k^* = 340$. This leaves 60 clauses per class and 360 in total, a reduction of 85%. At that budget, the pruned TM achieves macro-F1 0.6615 ± 0.0115 against 0.6660 ± 0.0125 for the full pool (Table I). Pruning therefore removes most of the pool with a mean macro-F1 decrease of 0.0045.

Table III shows both evidence signs at each pool’s selected setting for seed 42. Summed positive/negative supports are 150/585 for the full pool and 185/233 after pruning. These CAS signs are estimated from the pooled clauses across all six classes and differ from the original TM polarities. Shared clauses are counted in each class. The selected thresholds differ, so pruning need not reduce the summed supports.

IV. CONCLUSION

In this paper, we introduced CAS for compact, signed clause-based attribution of IoMT attacks. By combining weight-based pruning with complementary-pairs stability selection, CAS reduces a 2,400-clause TM to 216 distinct signed clauses while matching its macro-F1 of 0.6660. Further increasing the selection threshold reduces the representation to 193 clauses, with macro-F1 decreasing slightly to 0.6599. These results show that substantial clause reduction is possible with little loss in predictive performance in this application.

Ablations further support the use of conjunctive clauses and signed evidence. The theoretical results for pruning and stability selection hold under their stated assumptions, while further evaluation on additional datasets is needed to assess generalization to other applications.

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